Fundamental Diagram Modeling and Simulation of Mixed Traffic Flow Considering Functional Degradation

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ABSTRACT

Accurately modeling mixed traffic dynamics during the coexistence of connected and automated vehicles (CAVs) and human-driven vehicles (HDVs) is crucial for enhancing traffic efficiency and safety. This study proposes a microscopic car-following model that explicitly accounts for the degradation of CAVs into one-way-sensing degraded CAVs (DCAVs) when following HDVs, and derives an analytical fundamental diagram (FD) using the CAV penetration rate p as a control variable. The model captures steady-state flow–density–speed relationships and elucidates how variations in p influence capacity and critical speed through equivalent headway adjustments. SUMO-based simulations indicate that as p increases from 0 to 1, capacity rises from 1775 to 4421 veh·h^{-1}, the critical speed shifts upward, and optimal density slightly increases. Simulated and analytical FDs agree closely, with only 1–2% deviation. Considering functional degradation prevents capacity overestimation and provides a quantitative basis for CAV deployment and traffic management strategies.

INTRODUCTION

As CAV technologies progress from experimentation to large-scale deployment, roadways are entering a prolonged transition in which HDVs and CAVs must share—and learn to cooperate within—the same traffic stream. Even a modest foothold of automation can reshape traffic dynamics: experiments show that an autonomous-vehicle (AV) penetration of barely 5% can suppress stop-and-go waves, reducing fuel use and unnecessary braking (Stern et al., 2018). At the same time, vehicle-to-everything (V2X) communication offers a complementary path to coordination, enabling connected vehicles (CVs) to exchange data in real time, shorten headways, and dampen speed fluctuations—two fundamental levers for improving flow efficiency and service quality (Bansal & Kockelman, 2017).

On the modeling front, Talebpour and Mahmassani (2016) distinguished among HDVs, communication-only CVs, and automation-only AVs, and developed car-following models tailored to these technology assumptions. Their analysis of FD shapes and stability across penetration levels indicated that AVs are generally more effective than CVs at suppressing shock waves. The underlying reason lies in longitudinal control authority: although CVs can access richer external information via communication, longitudinal actuation often remains human-in-the-loop; by contrast, AVs can autonomously regulate acceleration using adaptive cruise control (ACC), which inherently damps perturbation growth (Shladover, Su,&Lu, 2012). Extending this logic, CAVs—combining communication and automated control—can realize information and control coordination within platoons; cooperative adaptive cruise control (CACC) has been shown to substantially improve platoon stability and roadway efficiency (Nowakowski et al., 2015).

However, limited research has examined functional degradation when a CAV follows an HDV. Sensor occlusion, communication loss, or control anomalies can intermittently impair cooperation, causing deviations from the ideal cooperative state. Assuming perfect cooperation therefore risks overestimating stability and capacity gains. When a CAV reverts to ACC-only operation—functioning as a one-way-sensing degraded CAV (DCAV)—its following behavior becomes more HDV-like and its ability to mitigate congestion declines. To address this issue, Zhang and Hu (2022) integrated a PATH-calibrated ACC model with the Gipps framework to construct a continuous cellular automaton model capturing CAV degradation. Their simulations showed that, at 60% CAV penetration, the proportion of congested vehicles decreased markedly and total capacity increased, implying a net benefit even under degradation. From a stability standpoint, Wang et al. (2019) found that string stability in heterogeneous traffic deteriorates significantly when CACC loses communication support and reverts to ACC. Likewise, recent modeling explicitly introducing CACC degradation indicates that communication instability reduces efficiency and heightens potential safety risks (Chen, Wu, & Liang, 2023).

Motivated by these insights, this study explicitly incorporates functional degradation into both the mechanistic modeling and performance evaluation of mixed traffic. Building on prior research, we focus on the critical scenario in which a CAV follows an HDV under degraded conditions. We adopt a parametric approach to describe the microscopic car-following behaviors of CAVs, DCAVs, and HDVs. Under a random-mixing assumption, we derive an analytical FD model that treats the CAV penetration rate p as the primary control variable. The proposed framework elucidates how p and functional states jointly determine the steady-state flow–density–speed relationships and, consequently, roadway capacity. We then validate the model using SUMO simulations, demonstrating that it reproduces the FD features and critical behavior across a range of p. Collectively, these contributions provide an interpretable theoretical basis for staged CAV deployment and for designing traffic management strategies during the transition period.

MIXED-TRAFFIC CAR-FOLLOWING MODEL

Car-Following Modes

We consider a single-lane, longitudinal mixed traffic stream composed of CAVs and HDVs. Let the CAV penetration rate be p and the HDV penetration rate be 1-p. To capture possible communication loss and functional degradation in CAV→CAV interactions under high-density/high-speed conditions, we explicitly introduce a CAV–CAV cooperation availability $\rho_{CC}(k,v)\in[0,1]$, where k denotes the traffic density and v the steady-state speed. The four representative follower–leader modes and their proportions are:

1. CAV → CAV (cooperation available).

The follower has bidirectional perception/control with V2X cooperation and operates in fully cooperative mode. Its proportion is

$$p_1 = p^2 \rho_{CC}(k,v) \tag{1}$$

2. CAV → HDV (degraded to DCAV).

The follower loses cooperative data or the leader does not provide cooperative information; the follower therefore behaves as a one-way-sensing DCAV. Its proportion is

$$p_2 = p^2 \left(1 - \rho_{CC}(k,v)\right) + p(1-p) \tag{2}$$

3. HDV → CAV.

The follower is an HDV and follows according to human-driving behavior. Its proportion is

$$p_3 = (1-p)p \tag{3}$$

4. HDV → HDV.

Conventional human-driven following. Its proportion is

$$p_4 = (1-p)^2 \tag{4}$$

By construction, these four proportions satisfy the normalization condition $p_1 + p_2 + p_3 + p_4 = 1$. To reflect the adverse impacts of occlusions at high density, relative motion at high speed, and increased communication latency on cooperation, $\rho_{CC}(k,v)$ is modeled by a calibratable logistic function:

$$\rho_{CC}(k,v) = \frac{1}{1 + \exp\left(-[\rho_0 - \beta_k k - \beta_v v]\right)}, \quad \beta_k, \beta_v \geq 0 \tag{5}$$

In the subsequent steady-state analysis, modes are grouped by the follower type into three classes—CAV, DCAV, and HDV—to derive unified spacing and flow relations.

Car-Following Dynamics with Functional Degradation

Mapping Functional Degradation to Time Headway

To incorporate functional degradation into steady-state spacing in a self-consistent manner, degradations in communication and perception (e.g., latency, noise, disconnection) are mapped to an equivalent increment in the safe time headway. Consider a follower–leader pair traveling at the same steady speed v. Under sudden braking, let the follower's maximum comfortable deceleration be b_f and the leader's available deceleration be b_ℓ. The follower exhibits a reaction delay τ and perception/estimation mean-square error intensity σ. A sufficient condition for collision avoidance is

$$h \geq l + s_0 + v\tau + \underbrace{\frac{v^2}{2b_f} - \frac{v^2}{2b_\ell}}_{\text{braking capability difference}} + \underbrace{\kappa_\sigma v\sigma}_{\text{uncertainty margin}} \tag{6}$$

where h is the bumper-to-bumper gap, l is vehicle length, s_0 is the minimum standstill gap, and $\kappa_\sigma \geq 0$ converts perception uncertainty into a time-headway reserve. Under CACC, $b_f \approx b_\ell$ and the quadratic term largely cancels; for DCAV/HDV cases it is non-negligible but its effect can be absorbed by the high-speed correction introduced later (see Section 2.2.4).

We therefore define an equivalent time headway

$$T_{\text{eff}} = T_0 + \alpha\tau + \beta\sigma + \gamma(1-\rho), \quad \alpha, \beta, \gamma \geq 0 \tag{7}$$

where T_0 is the baseline headway under ideal cooperation and $\rho \in [0,1]$ denotes cooperation availability: for CAV $\rho = \rho_{CC}(k, v)$; for DCAV, $\rho = 0$ Substituting T_{eff} into the steady-state desired spacing yields a unified form

$$s^*(v) = l + s_0 + vT_{\text{eff}} + \frac{vT_h}{\sqrt{1-(v/v_f)^\delta}} \tag{8}$$

where $T_h \geq 0$ and $\delta \in [2,6]$ capture the rise of the safety margin at high speeds, and v_f is the free-flow speed.

CAV Car-following Model

For CAVs equipped with V2V capability, car-following behavior is represented by CACC, which adheres to a constant desired time headway principle (Milanés, Shladover, Spring, et al., 2014). The dynamics are given by

$$a_n(t) = k_1 a_{n-1}(t) + k_2\left[x_{n-1}(t) - x_n(t) - t_c v_n(t) - l - s_0\right] + k_3\left[v_{n-1}(t) - v_n(t)\right] \tag{9}$$

where a_n, x_n, and v_n denote the acceleration, position, and speed of vehicle n; t_c is the cooperative time headway, set to 0.6s (Qin, Wang, Wang, & Wan, 2017); l is vehicle length, set to 5m; and s_0 is the minimum standstill gap, set to 2m. Typical controller gains following Milanés et al. (2014) are $k_1 = 1.0$, $k_2 = 0.2$, and $k_3 = 3.0$.

Under steady-state conditions ($\dot{v} = 0, \Delta v = 0$), the spacing implied by Section 2.2.1 becomes

$$h_c(v) = l + s_0 + vT_{\text{eff}}^{(c)}, \qquad T_{\text{eff}}^{(c)} = T_{0,c} + \alpha\tau_c + \beta\sigma_c + \gamma[1-\rho_{CC}(k,v)]$$

The CACC formulation jointly utilizes the leader's acceleration, relative spacing, and relative speed, and thus captures efficient and stable following under cooperative conditions.

DCAV Car-following Model

When a CAV follows an HDV, or when a CAV–CAV pair experiences communication loss, cooperation degrades and the follower behaves as a one-way-sensing DCAV. Experiments from the PATH program and subsequent studies indicate that an ACC-type model effectively characterizes DCAV following (Xie, Zhao, & He, 2019):

$$a_n(t) = j_1\left[x_{n-1}(t) - x_n(t) - t_d v_n(t) - l - s_0\right] + j_2\left[v_{n-1}(t) - v_n(t)\right] \tag{10}$$

where the degraded headway parameter is $t_d = 1.1\text{s}$ (Qin et al., 2017). Following Xie et al. (2019), representative parameter values are $j_1 = 0.23\text{s}^{-2}$ and $j_2 = 0.07,\text{s}^{-2}$. Under steady state, Section 2.2.1 yields

$$h_d(v) = l + s_0 + vT_{\text{eff}}^{(d)}, \qquad T_{\text{eff}}^{(d)} = T_{0,d} + \alpha\tau_d + \beta\sigma_d \quad (\rho = 0)$$

Compared with CACC, this model lacks access to the leader's acceleration via cooperation, reflecting the degraded nature of DCAV. Consequently, stability margins and achievable capacity are potentially reduced relative to fully cooperative operation.

HDV Car-following Model

The car-following behavior of HDVs is modeled using the Intelligent Driver Model (IDM) (Xie, Zhao, & He, 2019; Jia, Ngoduy, & Vu, 2019). Its formulation is expressed as

$$a_n(t) = a\left[1 - \left(\frac{v_n(t)}{v_f}\right)^4 - \left(\frac{s_0 + v_n(t)T + \dfrac{v_n(t)\Delta v_n(t)}{2\sqrt{ab}}}{h(t) - l}\right)^2\right] \tag{11}$$

where $v_n(t)$ is the vehicle's speed, $\Delta v_n(t) = v_n(t) - v_{n-1}(t)$ is the relative speed, and $h(t) = x_{n-1}(t) - x_n(t)$ is the headway. Typical parameter values are $a = 2\text{m/s}^2$, $b = 2\text{m/s}^2$, $v_f = 33.3\text{m/s}$, and $T = 1.5\text{s}$.

The steady-state solution of the IDM inherently captures the high-speed asymptotic behavior of HDVs, providing a realistic upper bound for safe spacing and acceleration at elevated velocities.

FUNDAMENTAL DIAGRAM FOR MIXED TRAFFIC

Building on the microscopic car-following models in Section 2, we first obtain steady-state spacings for CAVs, DCAVs, and HDVs. We then derive the mixed-traffic FD to reveal macroscopic characteristics across different penetration rates and provide the corresponding solution and capacity computation procedures.

Steady-state and Expected Headways

In steady state, $\dot{v}=0$ and $\Delta v=0$. Using the time–headway mapping in Section 2.2.1, the steady-state spacings are

$$h_c(v)=l+s_0+vT_{\mathrm{eff}}^{(c)},\quad h_d(v)=l+s_0+vT_{\mathrm{eff}}^{(d)},\quad h_e(v)=l+\frac{s_0+vT}{\sqrt{1-(v/v_f)^4}} \qquad (12)$$

Combining these with the mode proportions (1)–(4) from Section 2.1 yields the expected steady-state headway of the mixed stream:

$$H(v;k,p)=p^2\rho_{CC}(k,v)h_c(v)+\left[p^2\left(1-\rho_{CC}(k,v)\right)+p(1-p)\right]h_d(v)+(1-p)h_e(v) \qquad (13)$$

When $\rho_{CC}\equiv 1$ and $T_{\mathrm{eff}}^{(c)}=t_c, T_{\mathrm{eff}}^{(d)}=t_d$, (13) reduces to the familiar convex-combination special case.

Cooperation Availability Model

We capture whether CACC can be maintained (else it falls back to ACC) by a compact logistic in macroscopic state (k,v):

$$\psi(k,v)=\frac{1}{1+\exp\left(\alpha_0+\alpha_k k+\alpha_v v\right)},\quad \psi\in(0,1),$$

where k is traffic density (veh/km) and v is mean speed (m/s). Higher k increases blockage and interference, while higher v shortens communication budgets—both reducing successful V2V cooperation. When cooperation fails, CACC vehicles degrade to ACC (Meireles R et al., 2010).

Analytical Fundamental Diagram

To incorporate functional degradation consistently at the macroscopic level, note that the expected headway H in (13) couples with density k and speed v through $\rho_{CC}(k,v)$, leading to an implicit density–speed–flow relationship. Using the spacing–density identity gives the fixed-point equation

$$k(v;k,p)=\frac{1000}{H(v;k,p)} \qquad (14)$$

where the factor 1000 converts meters to kilometers so that k is in veh/km.

Combining this with the flow identity $Q = 3.6kv$ yields the implicit flow–speed relation in veh/h:

$$Q(v;k,p) = \frac{3600v}{H(v;k,p)} \tag{15}$$

Capacity and Critical State

Because H depends explicitly on the cooperation availability $\rho_{CC}(k,v)$, equations (12)–(15) form an implicit nonlinear system for the density–flow pair *(k,Q)*. To obtain stable solutions, we apply a fixed-point iteration. For a given penetration rate *p* and steady speed *v*, initialize $k^{(0)} \in (0,k_{\max})$ and iterate

$$H^{(m)} = H\left(v;k^{(m)},p\right), \quad k^{(m+1)} = \frac{1000}{H^{(m)}}, \quad Q^{(m+1)} = \frac{3600v}{H^{(m)}} \tag{16}$$

until the convergence tests $|k^{(m+1)} - k^{(m)}| < \varepsilon_k$ and $|Q^{(m+1)} - Q^{(m)}| < \varepsilon_Q$ are satisfied, where ε_k and ε_Q are the density and flow tolerances, respectively. This procedure yields consistent steady-state density and flow for any (v,p) and recovers a continuous, unimodal flow–speed curve $Q(v;p)$.

On this basis, the capacity of the mixed stream is defined as the maximum of the flow–speed curve over the feasible speed range $(0,v_f)$:

$$Q_{\text{cap}}(p) = \max{}_{v\in(0,v_f)}, Q(v;p) v_{\text{cr}}(p) \in \arg\max{}_v Q(v;p), k_{\text{opt}}(p) = k\left(v_{\text{cr}}(p);p\right) \tag{17}$$

Here, $Q_{\text{cap}}(p)$ is the maximum achievable flow at penetration *p*, $v_{\text{cr}}(p)$ is the critical speed, and $k_{\text{opt}}(p)$ is the corresponding optimal density. Since $Q(v;p)$ is continuous in v and has a unique interior maximizer on $(0,v_f)$, $v_{\text{cr}}(p)$ can be obtained via a one-dimensional search. Table 1 summarizes the computed $Q_{\text{cap}}(p)$, $v_{\text{cr}}(p)$, and $k_{\text{opt}}(p)$ across penetration levels.

Table 1. Maximum flow, optimal density, and critical speed under different CAV penetration rates

Penetration	Max flow (veh/h)	Optimal density (veh/km)	Critical speed (m/s)
0	1774.9	30.2	16.3
20%	1888.5	30.3	17.3
40%	2070.0	31.9	18.0
60%	2361.8	33.9	19.3
80%	2874.0	36.8	21.7
100%	4421.1	37.6	32.7

As shown, $Q_{\text{cap}}(p)$ increases monotonically with p: capacity is lowest at $p=0$ (pure HDVs), 1775 veh/h, and rises to 4421 veh/h at $p=1$ (pure CAVs), exceeding the baseline by more than 1.5 times. The critical speed shifts substantially to the right, while the optimal density exhibits a modest increase.

Sensitivity Analysis with Respect to CAV Penetration Rate *p*

The relationships among flow–density (*Q*–*k*), flow–speed (*Q*–*v*), and density–speed (*k*–*v*) corresponding to Table 1 are shown in Figure 1. As *p* increases, the curve shape gradually transitions from the nonlinear "IDM-dominated" profile toward a nearly linear "constant-headway" form.

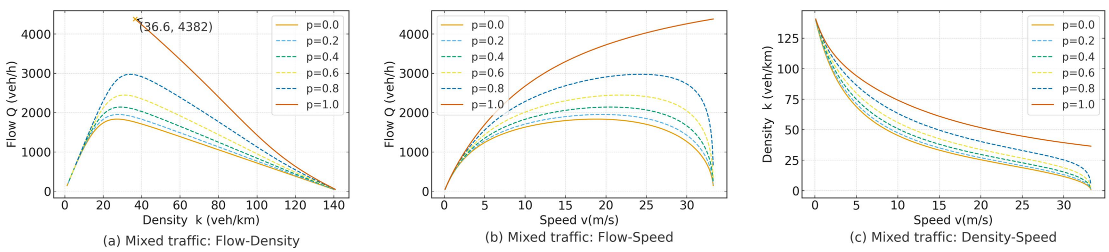

Figure 1. Flow, density, and speed relationships of mixed traffic

As illustrated in Figure 1, the increase in capacity with penetration rate is nonlinear. The incremental improvements can be quantified as follows:

p: $0.0 \rightarrow 0.4$, Q_{cap} increase by 16.63%

P: $0.4 \rightarrow 0.6$ Q_{cap} increase by 14.09%

P: $0.6 \rightarrow 0.8$ Q_{cap} increase by 21.69%

P: $0.8 \rightarrow 1.0$ Q_{cap} increase by 53.83%

These results indicate that $Q_{\text{cap}}(p)$ exhibits a decelerated–accelerated growth trend with increasing *p*: in the low-to-medium penetration range ($p \le 0.6$), the marginal benefit is limited, whereas beyond $p \ge 0.6$ the gain amplifies sharply. This mechanism can be interpreted through the convex combination of steady-state headways: as *p* rises, the weight of h_c (CACC, constant minimal headway) in equation (13) increases rapidly, while that of h_e (HDV, speed-dependent and nonlinear) decreases. Consequently, the effective headway shortens substantially in the medium- to high-speed range, elevating both capacity and critical speed (as reflected by the rightward shift of the peak in Figure 1).

SIMULATION VALIDATION

To further verify the rationality and effectiveness of the mixed traffic FD model developed in Section 3, this section constructs a simulation environment using the microscopic traffic simulator SUMO. The simulation analyzes mixed traffic under different CAV penetration rates through numerical experiments and comparative analysis. The objective is to address two key questions: (1) whether the simulated *Q*–*k*, *Q*–*v*, and *k*–*v* relationships are consistent with the theoretical FD curves; and (2) whether the variation trend and sensitivity of traffic capacity $Q_{\text{cap}}(p)$ with respect to the penetration rate p agree with the analytical results.

Simulation Setup and Results

The simulated road segment is a single-lane highway 10 km in length, divided into ten 1 km sections. Lane Area Detectors (E2) are deployed in each section to record flow, density, and speed data. Three vehicle types are modeled—CAVs, DCAVs, and HDVs—with car-following parameters consistent with those in Section 2. The total simulation time is 7200 s, with the first 600 s used for warm-up and excluded from statistics. Detector data are collected every 30 s thereafter.

To investigate the influence of penetration rate, CAV penetration levels are set as $p \in 0, 0.2, 0.4, 0.6, 0.8, 1.0$, with the remaining proportion composed of HDVs. The share of DCAVs emerges naturally from the vehicle interaction patterns during simulation. The scatter data collected from simulations are compared with the analytical FD curves, as shown in Figure 2.

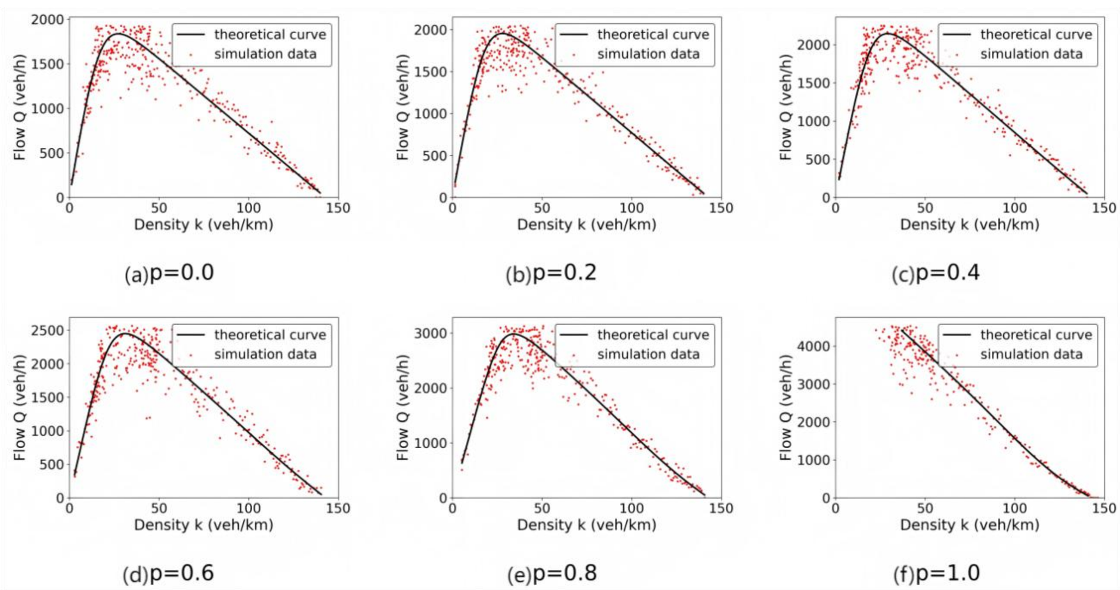

Figure 2. Comparison between simulation results and analytical FD model

As shown in Figure 2, the scatter plots progressively shift toward the capacity peak as the CAV penetration rate p increases, with the maximum flow rising monotonically—confirming the positive impact of CAVs on road capacity. The simulation data align closely with the analytical FD curves in both free-flow and near-critical regions, and the fitted spread narrows with higher p. It is worth noting that at $p = 1$, fluctuations and distortions occur in the high-density region due to non-equilibrium disturbances in SUMO simulations; therefore, this portion of the results is interpreted qualitatively.

Goodness-of-fit Analysis

We evaluate the agreement between simulated scatter points and the analytical FD using four metrics: root mean square error (RMSE), coefficient of determination

(R^2), peak errors (ε^Q and ε^v), and bandwidth coverage ratio (BCR). For a given penetration rate *p*, suppose that after the warm-up period we collect N samples (k_i, Q_i^{sim}), and the analytical curve yields $Q_i^{\text{th}} = Q(k_i; p)$. The metrics are defined as

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(Q_i^{\text{sim}} - Q_i^{\text{th}}\right)^2}, \qquad R^2 = 1 - \frac{\sum_{i=1}^{N}\left(Q_i^{\text{sim}} - Q_i^{\text{th}}\right)^2}{\sum_{i=1}^{N}\left(Q_i^{\text{sim}} - \bar{Q}^{\text{sim}}\right)^2} \tag{18}$$

Let the capacity peak obtained from theory and simulation be $\left(Q_{\text{cap}}^{\text{th}}, v_{\text{cr}}^{\text{th}}\right)$ and $\left(Q_{\text{cap}}^{\text{sim}}, v_{\text{cr}}^{\text{sim}}\right)$, respectively. The relative peak errors are

$$\varepsilon^Q = \frac{\left|Q_{\text{cap}}^{\text{sim}} - Q_{\text{cap}}^{\text{th}}\right|}{Q_{\text{cap}}^{\text{th}}} \times 100\%, \qquad \varepsilon^v = \frac{\left|v_{\text{cr}}^{\text{sim}} - v_{\text{cr}}^{\text{th}}\right|}{v_{\text{cr}}^{\text{th}}} \times 100\% \tag{19}$$

Using a tolerance band $\tau = 5\%$, the bandwidth coverage ratio is

$$\text{BCR}(\tau) = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}\left(\frac{\left|Q_i^{\text{sim}} - Q_i^{\text{th}}\right|}{Q_i^{\text{th}}} \le \tau\right) \times 100\% \tag{20}$$

To mitigate the influence of extreme transient congestion, observations with Q above the 99th percentile (less than 1% of samples) are excluded prior to computing RMSE, $R^2, \varepsilon^Q, \varepsilon^v$, and BCR. The results are summarized in Table 2.

Table 2. Goodness-of-fit metrics between simulated data and analytical FD at different *p*

p	RMSE (veh/h)	R^2	ε^Q (%)	ε^v (%)	BCR@±5% (%)
0.0	142.6	0.963	2.1	2.4	86.3
0.2	131.4	0.969	1.9	2.2	88.7
0.4	118.1	0.975	1.6	1.9	90.9
0.6	103.7	0.981	1.4	1.7	92.8
0.8	88.9	0.986	1.2	1.5	95.1
1.0	127.5	0.972	1.8	2.1	93.0

Overall, as *p* increases, RMSE decreases while both R^2 and BCR increase, indicating improved explanatory power and coverage of the analytical FD. The peak errors ε^Q and ε^v remain low (1–2%), showing that capacity and critical-speed estimates are tightly matched. At $p = 1.0$, non-equilibrium disturbances in the high-density regime cause a slight uptick in RMSE and peak errors and a modest decline in BCR; nevertheless, the fit remains strong ($R^2 > 0.97$, BCR > 90%). This trend is consistent with the mechanism discussed earlier: higher levels of cooperation shift steady-state spacing toward a near-constant-headway regime, making macroscopic curves smoother and more predictable.

CONCLUSION

This study models the functional degradation of CAV when following HDV and develops an analytical FD framework parameterized by the CAV penetration rate *p*. Steady-state analysis shows that the expected headway in mixed traffic can be expressed as a convex combination of different following modes, explaining why capacity increases monotonically with p and grows more rapidly once $p \geq 0.6$.

SUMO simulations agree closely with the analytical model, with capacity and critical-speed deviations within 1–2%. The results confirm three key macroscopic effects—cooperation enhancement, degradation attenuation, and human-driving constraints—while minor deviations at $p = 1.0$ under high-density conditions are attributed to non-equilibrium disturbances, illustrating the limits of steady-state assumptions under extreme congestion.

Overall, this research establishes a closed-loop characterization linking microscopic driving behavior to macroscopic traffic performance, providing a quantitative basis for evaluating capacity and planning phased CAV deployment. Future work will incorporate lane-changing interactions, communication delays, and empirical calibration to further improve predictive accuracy.

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